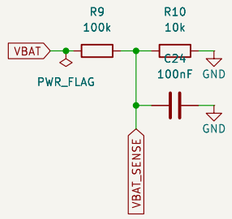
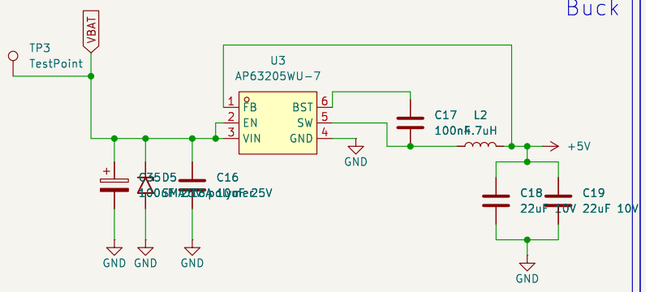
Power

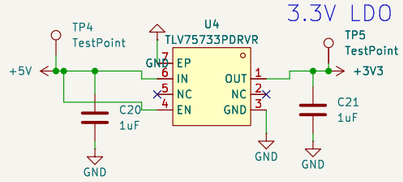
Battery sense divider



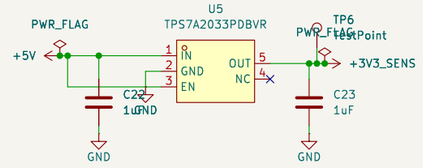
Buck



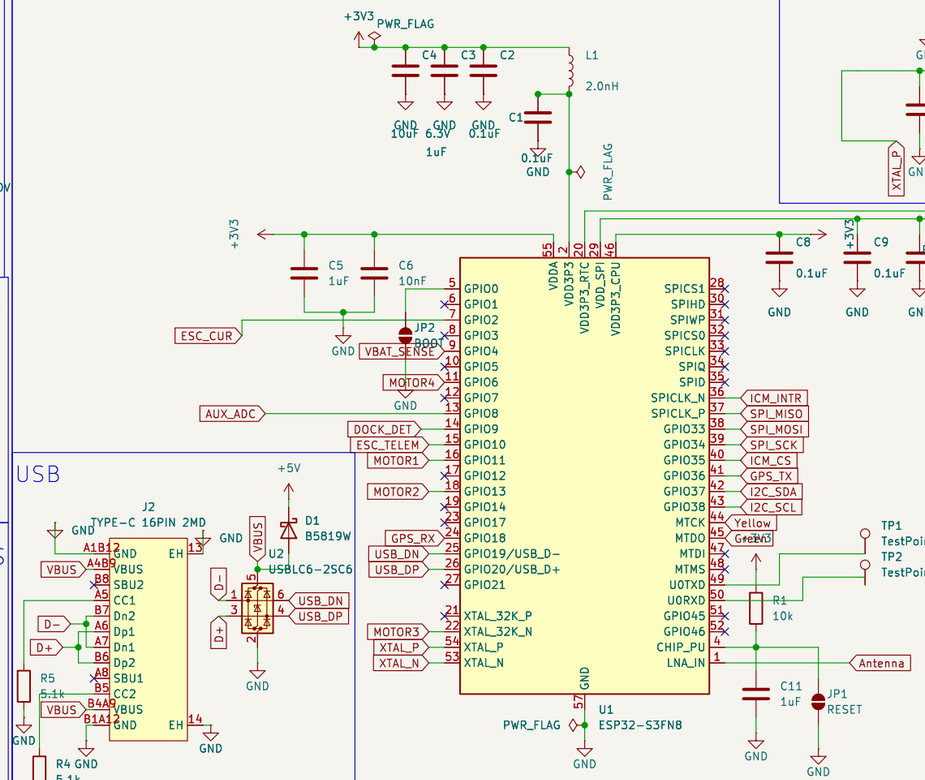
3.3V LDO – ESP, LEDs



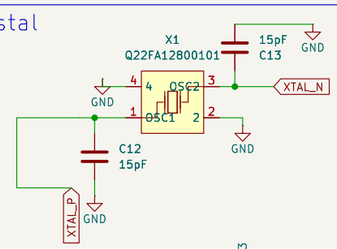
3.3V LDO IMU, BAROMETER, GPS



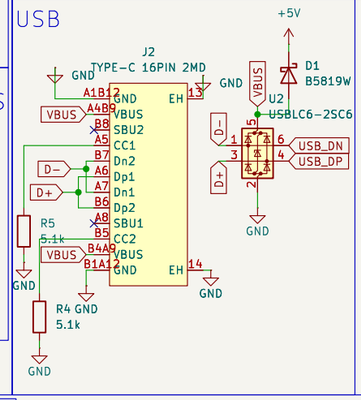
MCU



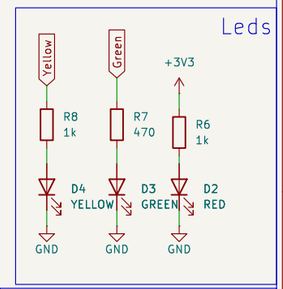
Crystal



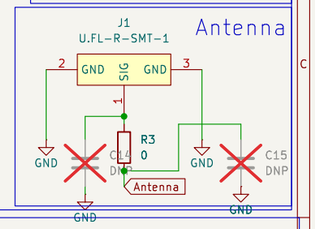
USB



Leds

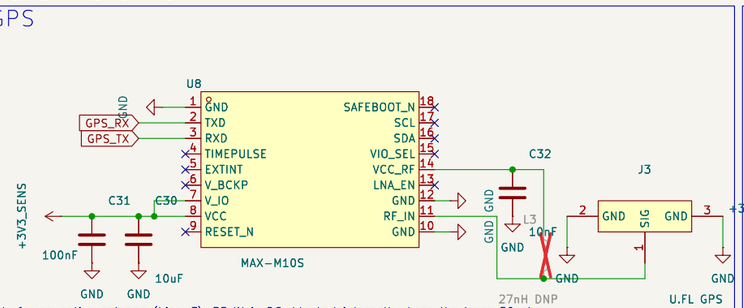


Antenna



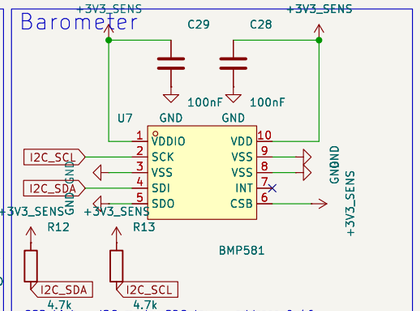
Sensors

GPS



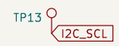
L3 only for an active antenna (bias-T). RF_IN is DC-blocked internally; keep the trace 50 ohm.

Barometer



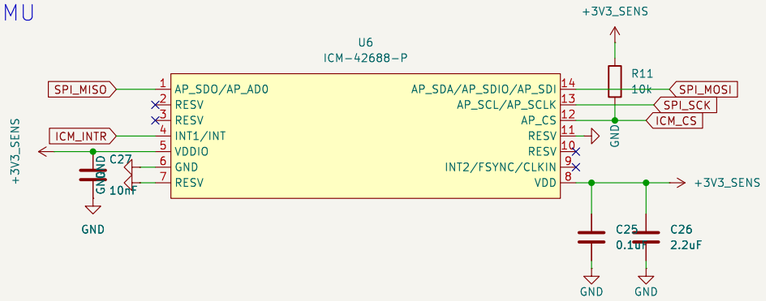
CSB high = I2C mode, SDO low = address 0x46

Test points



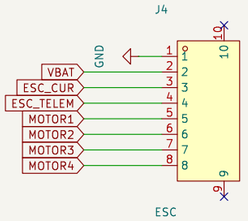
Bottom-side pads for probing the buses and recovering the ESP32.\nShort the RESET or BOOT pad pair to GND with tweezers.

IMU



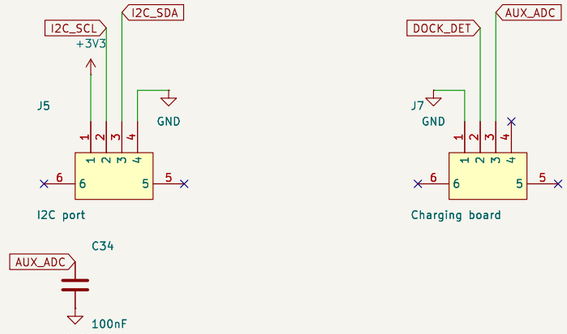
Connectors

ESC (JST-SH 8)



Pin order follows the Betaflight 4-in-1 standard:
GND, VBAT, CUR, TLM, M1–M4. Confirm against the ESC.

Ports (JST-GH 4)



I2C port uses the Dronecode order: VCC, SCL, SDA, GND.
Charging port: GND, DOCK_DET, AUX_ADC, spare.

Texas Aerial Robotics – Team 1

Sheet: /
File: network-drone-fc.kicad_sch

Title: Network Drone Flight Controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 1/1